

# Practical 55 — Algorithm: Login Validation System

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## 1. Overview & Scenario

Almost every software application requires users to log in before accessing features. To improve code readability and maintainability, software companies separate the login validation logic into dedicated functions.

## 2. Functional Requirements

- Store a predefined username and password.
- Accept login credentials from the user.
- Validate the credentials using a separate function `validate_Login()`.
- Display an appropriate message (Login Successful / Welcome Administrator! or error message).
- Lock the account after three unsuccessful login attempts.

## 3. Algorithm Specifications

### Main Program Algorithm

1. **START**
2. Define constants for `STORED_USERNAME` ("admin"), `STORED_PASSWORD` ("admin123"), and `MAX_ATTEMPTS` (3).
3. Initialize loop variables: `attempts_left = MAX_ATTEMPTS` and `is_logged_in = 0`.
4. **WHILE** `attempts_left > 0`:
  - Prompt user and `READ` `input_username`.
  - Prompt user and `READ` `input_password`.
  - Call function `validate_Login(input_username, input_password)`.
  - **IF** result is equal to 1 (True):
    - Set `is_logged_in = 1`.
    - Break/Exit loop.
  - **ELSE** (result is 0):

- Decrement attempts\_left by 1.
  - If attempts\_left > 0, display invalid credentials error and remaining attempts.
5. **IF** is\_logged\_in == 1:
    - PRINT "Login Successful"
    - PRINT "Welcome Administrator!"
  6. **ELSE:**
    - PRINT "Account Locked!"
    - PRINT "Maximum attempts reached."
  7. **STOP**

### **Validation Sub-Routine: validate\_Login(user, pass)**

1. **START SUB-ROUTINE**
2. Compare user with STORED\_USERNAME AND pass with STORED\_PASSWORD.
3. **IF** both strings match identically:
  - RETURN 1 (Valid)
4. **ELSE:**
  - RETURN 0 (Invalid)
5. **END SUB-ROUTINE**

## **4. Input / Output Specifications**

| Parameter            | Value                                      |
|----------------------|--------------------------------------------|
| <b>Sample Input</b>  | Username : admin<br>Password : admin123    |
| <b>Sample Output</b> | Login Successful<br>Welcome Administrator! |